

IRENE AGUSTI

☎ 641-260-4161 ✉ ireneagustimazza@gmail.com [in linkedin.com/in/irene-agusti](https://www.linkedin.com/in/irene-agusti) github.com/ireagusti02

Education

Bachelor of Arts in Mathematics

December 2024

Grinnell College

Grinnell, IA

- Relevant Coursework: Data Science, Machine Learning, Linear Algebra, Numerical Analysis, Partial Differential Equations, Applied Statistics, Probability and Statistics, Modern Physics, Abstract Algebra.

GPA: 3.70

Experience

Data Science Mentor

August 2024 - Present

Grinnell College, Data Science and Inquiry Lab

Grinnell, IA

- Provide guidance and mentorship in R, Python, MATLAB, and Tableau by assisting students with data analysis, visualization, and computational projects.
- Lead the development of an automated web scraping solution in Python for the athletic department, aimed at modernizing legacy data storage systems and designing an interactive dashboard for tracking school records and athlete performance metrics.

Time Series Machine Learning Researcher

May 2024 - August 2024

Grinnell Mathematics and Statistics Department

Grinnell, IA

- Developed a synthetic data generation method for semi-reflective lane deviation data in impaired driving detection, achieving a 0.05 AUC improvement using autoregressive models and random walks methods.
- Led design and execution of experiments on 450 augmented datasets, optimizing deep learning model performance (InceptionTime) through ROC AUC evaluation and employing a bias-reducing validation approach.
- Conducted comprehensive data cleaning and preparation, manipulating datasets for feature engineering, faceted t-tests, and visualizations in R, summarizing results in a 10-page research report.
- Summarized findings in a 10-page research report, which was peer-reviewed and recommended for presentation in the AI and Advanced Computing section at the TRB Annual Conference.

Microfinance Intern

May 2022 – July 2022

The Iowa Center for Economic Success

Des Moines, IA

- Compiled over 20 financial reports based on business' bank statements, taxes, and accounting sheets. Analyzing small local businesses' loan risk and providing necessary feedback on necessary financial adjustments.
- Developed a business plan creation tool and corresponding documentation to help automate the process, reducing the business plan-making process for new loan applicants by 15%.

Projects

Generative Semi-Reflective Data Augmentation (GSRDA) | Python

August 2024

- Implemented a generative data augmentation method aimed at improving impaired driving classification from limited driving simulator data. Resulting in 0.05 AUC improvement in high-speed curves data.

Numerical Methods Portfolio | MATLAB

December 2023

- Explicitly developed scripts and functions to perform well-known numerical methods including Newton's Method, Singular Value Decomposition, Adaptive Quadrature, Gauss-Seidel, and Newton Interpolation.

Tennis Court Finder | R Shiny

October 2023

- Developed an R Shiny app that filters court information based on desired characteristics such as location, surface, lights, access to pro-shop, number of courts, and accessibility. Refined the interface to create a user-friendly and eye-catching design.

Leadership and Extracurriculars

Varsity Tennis

2020 – 2024

Grinnell College

Grinnell, IA

- Served as Captain for the 2023-2024 season, leading the team to its highest-ever national ranking of 21st.
- Achieved individual national rankings of 16th in 2024 and 43rd in 2022.

Other Skills

Programming: MATLAB, Python, R.

Technical: Microsoft Office 365, L^AT_EX, Salesforce, FormTitan, Lenderfit, Minitab, Qualtrics, PowerBI, Tableau.

Languages: English, Spanish, Catalan, and Basic Russian.